

Electromagnetism 2

(spring semester 2026)

Lecture 7

Dielectric materials; polarisation

- ❖ Dielectric polarisation
- ❖ Surface and volume polarisation charges
- ❖ Electric susceptibility
- ❖ Microscopic theory of dielectric gases

Previous lecture

Maxwell's equations in free space (remember them)

$$(M1) \quad \nabla \vec{E} = \rho / \epsilon_0$$

$$(M2) \quad \nabla \vec{B} = 0$$

$$(M3) \quad \nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$(M4) \quad \nabla \times \vec{B} = \mu_0 \vec{j} + \epsilon_0 \mu_0 \frac{\partial \vec{E}}{\partial t}$$

Displacement current density: $\vec{j}_D = \epsilon_0 \frac{\partial \vec{E}}{\partial t}$

The continuity equation, $\nabla \cdot \vec{j} = - \frac{\partial \rho}{\partial t}$, follows from (M1) and (M4). 1

Conductors vs dielectrics

Conductors

Free charges

Currents can flow: $\mathbf{j} \neq 0$

Large conductivity σ [Ohm's law: $\mathbf{j} = \sigma \mathbf{E}$]

In the static limit, $\rho = \mathbf{E} = 0$ inside

Surface+volume are equipotential

Free charges on surface only

$\mathbf{E}$ is perpendicular to surface

Dielectrics (insulators)

Bound charges

No currents flow: $\mathbf{j} = 0$

Conductivity $\sigma = 0$

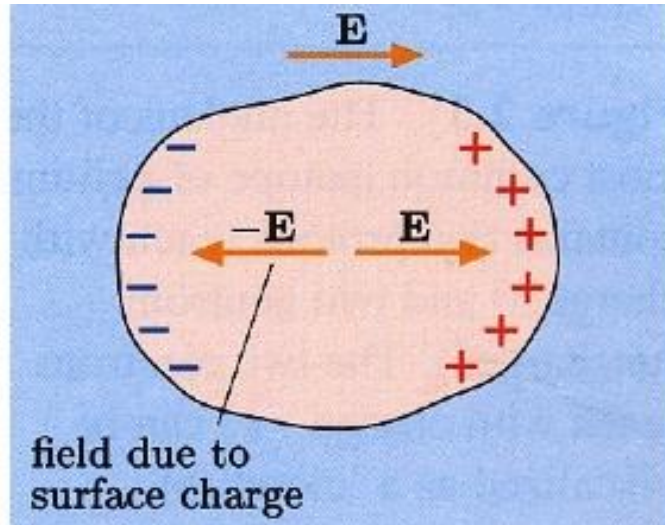
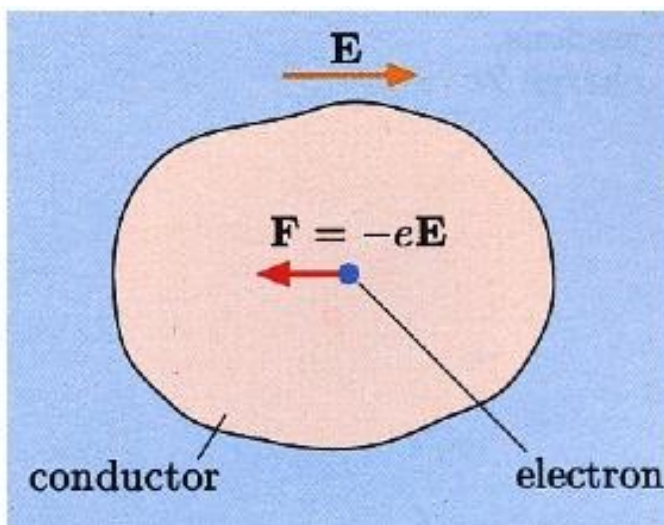
In the static limit, $\mathbf{E} \neq 0$ inside

Surface+volume are **not** equipotential

Bound charges can be anywhere

Any $\mathbf{E}$ allowed at surface

Free charges in conductors:



Dielectric polarisation

Main sources of **dielectric polarisation**:

- 1) **Electronic polarisation**: deformation of electronic clouds, alignment of **induced dipole moments** of the atoms along external **E**-field.
- 2) Orientation of **permanent molecular dipole moments** along the **E**-field (for polar molecules).

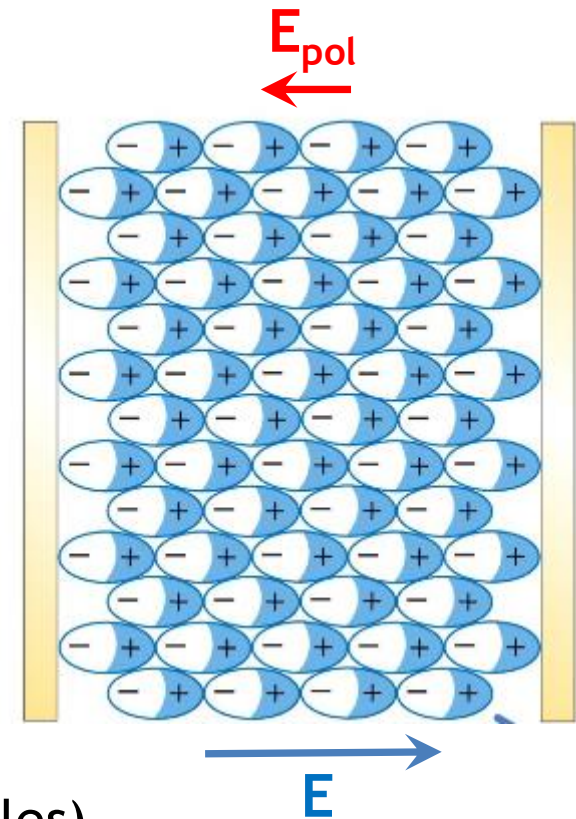
Polarisation **reduces** the field in the dielectric.

Polarisation vector: dipole moment induced per unit volume,

$$\vec{P} = \frac{1}{\Delta V} \sum_{\Delta V} \vec{p}_i \quad \begin{array}{l} \text{(sum over atoms/molecules)} \\ [\text{C/m}^2] \end{array}$$

Dielectric materials are attracted into regions of strong **E**-field: dipole momenta (**p**) are aligned with **E**-field;

force on a dipole in non-uniform field is $F_z = \mathbf{p} \cdot d\mathbf{E}/dz$ (lecture 4).



Surface polarisation charges

Polarisation gives rise to bound *surface charges*.

Dipole moment of a slice
(using the definition of the dipole moment):

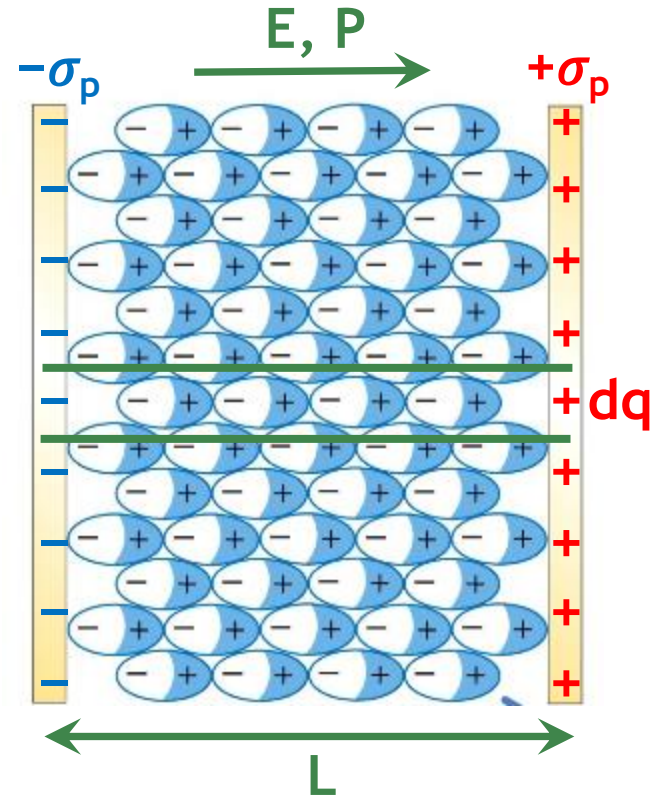
$$dp = dq \cdot L = \sigma_p dS \cdot L = \sigma_p dV$$

Total dipole moment of the sample:

$$p_{\text{total}} = \int_V dp = \int_V \sigma_p dV = \sigma_p V$$

Polarisation (using the definition):

$$P = p_{\text{total}}/V = \sigma_p$$



More generally, for a surface not perpendicular to the polarisation vector,

$$\sigma_p = \vec{P} \cdot \vec{n}$$

$\vec{n}$: unit normal vector pointing outwards.

No surface charges on the top and bottom surfaces in the sketch.

Volume polarisation charges

Elementary surface polarisation charge: $dQ = \vec{P} \cdot d\vec{S}$

Apply this to any volume V inside the dielectric to compute the *volume polarisation charge* density ρ_p :

$$Q_p = \int_V \rho_p dV = - \int_S \vec{P} d\vec{S} = - \int_V \nabla \cdot \vec{P} dV$$

Charge displaced
into the volume V

Divergence theorem (*lecture 2*)

Therefore $\rho_p = -\nabla \cdot \vec{P}$ (no volume charges for uniform polarisation)

If electric field $\mathbf{E}(t)$ varies with time, *polarisation current can* flow:

$$\nabla \cdot \vec{j}_p = -\frac{\partial \rho_p}{\partial t} = \frac{\partial}{\partial t} \nabla \cdot \vec{P}, \text{ therefore } \vec{j}_p = \frac{\partial \vec{P}}{\partial t}.$$

Continuity equation (*lecture 6*)

The distributions σ_p and ρ_p represent the effect of polarisation via equivalent macroscopic charge distributions.

Electric susceptibility

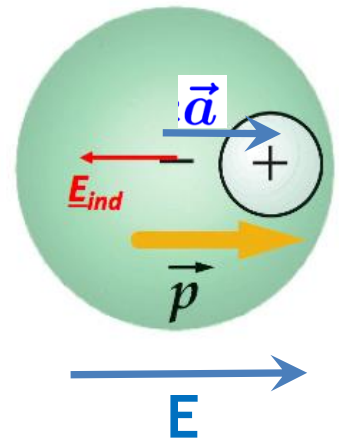
Consider a linear, isotropic and homogeneous (“LIH”) dielectric.

The *electric susceptibility* χ_E is defined as the (dimensionless) constant of proportionality between polarization and field:

$$\vec{P} = \chi_E \cdot \epsilon_0 \cdot \vec{E}$$

Should we expect linearity?

1) *Electronic polarisation*: electric field $\mathbf{E}$ causes displacement of electron cloud density wrt nucleus. Therefore, each atom becomes polarised.



Typical field “seen” by electrons: $E = \frac{e}{4\pi\epsilon_0 r_{\text{Bohr}}^2} \sim 10^{11} \text{ V/m}$
(huge, greatly exceeds any external field)

Therefore, the displacement is $a \ll r_{\text{Bohr}}$ (typically $\sim 1 \text{ fm}$), and depends *linearly* on $\mathbf{E}$.

Induced dipole moment

Atomic/molecular polarisability α [m³]:

property of atoms/molecules,

$$\vec{p} = \alpha \epsilon_0 \vec{E}_{\text{local}}$$

A crude estimate: $\alpha \approx 4\pi r_{\text{atom}}^3$ (see non-assessed problems).

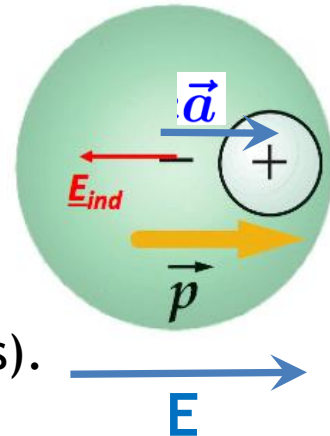
Electric susceptibility for a **gaseous non-polar dielectric**:

$$\chi_E = \frac{P}{\epsilon_0 E} = \frac{np}{\epsilon_0 E} = \frac{n\alpha\epsilon_0 E}{\epsilon_0 E} = n\alpha$$

α : molecular polarisability [m³];

n : density of molecules [m⁻³];

p : induced dipole momentum of a single molecule [C·m].

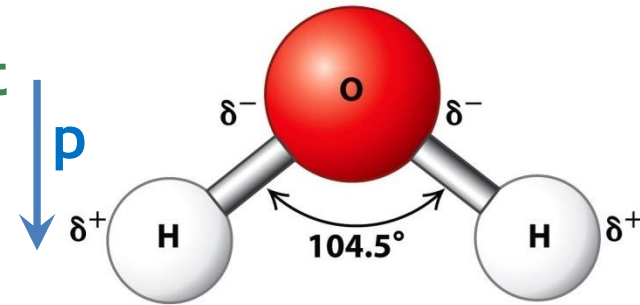


- ❖ Susceptibility depends on the density only; $\chi_E \sim 10^{-4}$ for gases at STP.
- ❖ Susceptibility does not depend on temperature at fixed density (assuming molecules do not raise to excited states or dissociate).

Polar molecules

2) *Polar molecules*: non-zero dipole moment

Example: water (H_2O).



Typical molecular dipole moment:

$$p \approx 1.6 \times 10^{-19} \text{ C} \cdot 10^{-10} \text{ m} \sim 10^{-29} \text{ Cm}$$

Even for a very strong field ($E=10^6 \text{ V/m}$),
the work required to revert the direction of p (*lecture 4*)

$$W = 2pE = 2 \times 10^{-29} \text{ Cm} \cdot 10^6 \text{ V/m} \sim 10^{-4} \text{ eV}$$

is much smaller than thermal motion energy at room temperature:

$$\frac{3}{2}kT \approx 0.04 \text{ eV}$$

Weak alignment of the dipoles, no saturation, *linearity*.

Orientation of dipole moments

In the absence of electric field, the density of dipole moments is isotropic:

$$dN = \frac{d\Omega}{4\pi} = \frac{2\pi \sin \theta d\theta}{4\pi} = \frac{1}{2} \sin \theta d\theta$$

In the presence of an external electric field,

$$dN = C \cdot \underbrace{\exp\left(-\frac{U}{kT}\right)}_{\text{Boltzmann factor}} \cdot 2\pi \sin \theta d\theta$$

θ : angle between dipole moment and field direction

$$\approx C \cdot \left(1 - \frac{U}{kT}\right) \cdot 2\pi \sin \theta d\theta$$

$$= C \cdot \left(1 + \frac{pE \cos \theta}{kT}\right) \cdot 2\pi \sin \theta d\theta$$

Normalisation: $\int_0^\pi dN = 4\pi C = 1$, therefore $C = \frac{1}{4\pi}$

Polarisation and susceptibility

Polarisation = dipole moment of a unit volume:

$$P = n \int_0^\pi \underbrace{\frac{1}{2} \sin \theta \left(1 + \frac{pE \cos \theta}{kT} \right)}_{\text{Density of dipole moments}} \cdot \underbrace{p \cos \theta}_{\text{Component along the field direction}} \cdot d\theta$$

$$= \frac{np^2 E}{2kT} \int_0^\pi \sin \theta \cos^2 \theta \cdot d\theta = -\frac{np^2 E}{2kT} \int_0^\pi \cos^2 \theta d(\cos \theta) = \frac{np^2 E}{3kT}$$

Net polarisation of gaseous polar dielectric:

$$\vec{P} = \left(n\alpha\epsilon_0 + \frac{np^2}{3kT} \right) \vec{E}$$

Electric susceptibility for a **dielectric gas**, $\chi_E = P/(\epsilon_0 E)$:

$$\chi_E = n \left(\alpha + \frac{p^2}{3\epsilon_0 kT} \right)$$

Susceptibility vs temperature

For a **dielectric gas**,

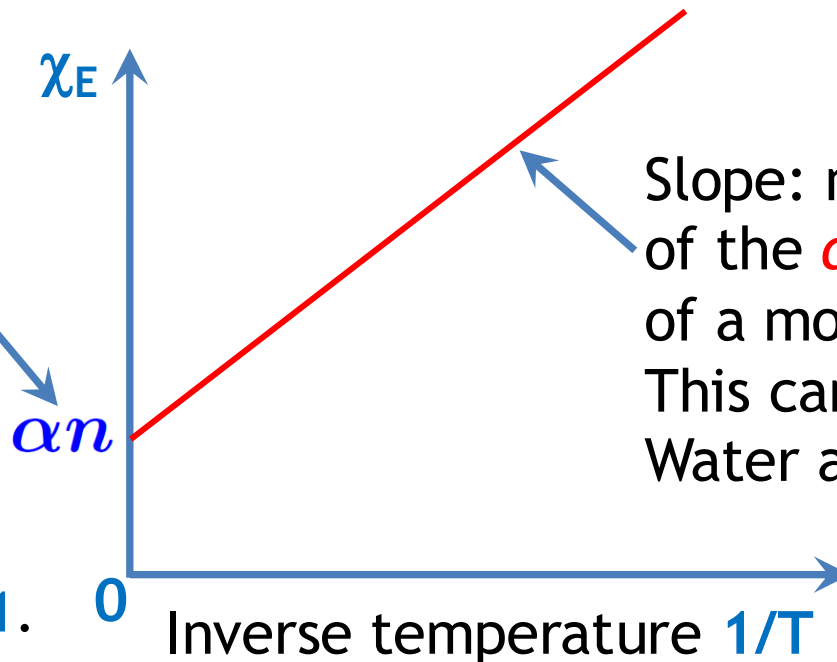
$$\chi_E = n \left(\alpha + \frac{p^2}{3\epsilon_0 kT} \right)$$

At a **fixed density** (i.e. **$n=\text{const}$**):

Intercept:
susceptibility
due to **electronic
polarisation**,
direct measurement
of the **molecular
polarisability**.

Monoatomic gases
at STP: **$\alpha n \sim 10^{-4}$** .

Solid dielectrics: **$\alpha n \sim 1$** .



Slope: measurement
of the **dipole moment**
of a molecule.

This can generate **$\chi_E \gg 1$** .
Water at **$T=300\text{K}$** : **$\chi_E=80$** .

Liquid and solid dielectrics

[Not discussed in the lecture]

Liquids and solids have much higher densities (n) than gases:

- ❖ Much larger electronic polarisation term, αn .
- ❖ The field on each individual atom is influenced by polarisation of the neighbouring atoms.
- ❖ As a result, further enhancement of electronic polarisation.
Clausius-Mossotti relation for non-polar liquid dielectrics:

$$\chi_E = \frac{n\alpha}{1 - \frac{1}{3}n\alpha}$$

- ❖ Solids (including piezoelectricity, pyroelectricity, ferroelectricity) and polar liquids are not discussed in this course.

Summary

- ❖ Polarisation of LHM dielectric materials, characterised by the electric susceptibility χ_E :

$$\vec{P} = \frac{1}{\Delta V} \sum_{\Delta V} \vec{p}_i = \chi_E \cdot \epsilon_0 \cdot \vec{E}$$

- ❖ Volume distribution of polarisation $\mathbf{P}$ is equivalent to the following distribution of surface and volume charges:

$$\sigma_p = \vec{P} \cdot \vec{n} \quad \text{and} \quad \rho_p = -\nabla \cdot \vec{P}$$

- ❖ Electric susceptibility for a dielectric gas:

$$\chi_E = n \left(\alpha + \frac{p^2}{3\epsilon_0 kT} \right)$$

The two terms are due to *electronic polarisation* and the *alignment of permanent dipole moments p* of molecules.